
Re: ["Prospective Undergraduate Student"] Interest in LLM Reasoning and Interpretability Research

From Ziyu Yao <ziyuyao@gmu.edu>

Date Mon 22-Jun-26 3:15 PM

To Mohan Krishna Vallabhaneni <mvallab@gmu.edu>

Hi Mohan,

Great! I just sent a Zoom invite for this Thu 2pm.

Best,
--Ziyu

From: Mohan Krishna Vallabhaneni <mvallab@gmu.edu>

Date: Sunday, June 21, 2026 at 9:14 PM

To: Ziyu Yao <ziyuyao@gmu.edu>

Subject: Re: ["Prospective Undergraduate Student"] Interest in LLM Reasoning and Interpretability Research

Hello Professor Yao,

I really appreciate in sharing the paper, I will definitely look into it.
And it is a really great news, I am really interested in having a meet with you and Thursday sounds great, whenever you are available to meet.

Best Regards,
Mohan Krishna Vallabhaneni.

From: Ziyu Yao <ziyuyao@gmu.edu>

Sent: Sunday, June 21, 2026 6:15 PM

To: Mohan Krishna Vallabhaneni <mvallab@gmu.edu>

Subject: Re: ["Prospective Undergraduate Student"] Interest in LLM Reasoning and Interpretability Research

Hi Mohan,

Sounds good! We have another recent paper looking into mechanisms for syntactic code completion (specifically, balanced parentheses completion):
https://proceedings.neurips.cc/paper_files/paper/2025/file/c0159f3dffb225f51d75fab412a99683-Paper-Conference.pdf.

Maybe let's have a meeting to meet with each other first? I'm working from home over the summer. I'm available most of the afternoons but for this week I prefer Wed and Thu 2pm. Let me know if this works for you.

Best,
--Ziyu

From: Mohan Krishna Vallabhaneni <mvallab@gmu.edu>
Date: Sunday, June 21, 2026 at 1:03 PM
To: Ziyu Yao <ziyuyao@gmu.edu>
Subject: Re: ["Prospective Undergraduate Student"] Interest in LLM Reasoning and Interpretability Research

Hello Professor Yao,

Thank you so much for your response. I apologize for my delayed reply. I wanted to take the time to properly read the paper you shared before responding.

I read "All for One: LLMs Solve Mental Math at the Last Token With Information Transferred From Other Tokens," and I found it genuinely interesting. My original question was about how the DCD idea could extend to more open-ended reasoning tasks, especially math reasoning and code generation. After reading this paper, I feel that the math-reasoning side of my question was answered much more clearly. The paper helped me understand that LLMs may solve reasoning tasks very differently from how humans naturally think about them.

One way I understood the paper was through a group-project analogy, which I thought of for myself. I imagined the tokens as students working on a math problem. During the waiting period, each student works independently and performs only general background processing without sharing their specific values. During the transfer period, the students pass their relevant notes to the last student. Then, in the final stage, the last student has the important information from the others and performs the final calculation. This analogy helped me understand the idea that meaningful computation can happen very late and mainly at the last token, after information has been transferred from other tokens.

I also found the methodology very engaging, especially how the paper starts from the transformer architecture and then carefully modifies the computation flow using Context-Aware Mean Ablation (CAMA) and Attention-Based Peeking (ABP). I liked how CAMA preserves context-aware background information while reducing input-specific computation early on, and how ABP controls where information transfer is allowed through attention. Even the naming detail of CAMA being related to "Camel" and "Llama" made the paper more memorable and fun to read and finally how only the last token of each layer can get the information from the previous one.

After reading this paper, I am even more excited about the code-generation side of the question. Since code generation is more open-ended and can involve syntax, logic, planning, debugging, and problem decomposition, I think it would be very interesting to explore whether LLMs internally organize code-generation tasks differently from how humans understand or categorize them. This direction feels like a natural extension of both the DCD paper and the arithmetic paper you shared.

I would be very happy to work with you on a well-scoped project along this direction. Could you please let me know how I should start preparing and what the next steps would be for working under your guidance? I would be happy to read more related papers, review the DCD setup, study any codebase, or meet with you to discuss the project scope and expectations.

Thank you again for considering me. I am very grateful for the opportunity and excited to learn from your guidance.

Best regards,
Mohan Krishna Vallabhaneni

From: Ziyu Yao <ziyuyao@gmu.edu>

Sent: Friday, June 19, 2026 11:53 PM

To: Mohan Krishna Vallabhaneni <mvallab@gmu.edu>

Subject: Re: ["Prospective Undergraduate Student"] Interest in LLM Reasoning and Interpretability Research

Hi Mohan,

Thank you for your enthusiastic email! I'm sorry for the delay during the summer when my response rate is typically slow.

I like the question you raised:

> One question I had while reading the paper is how this idea could extend to more open-ended reasoning tasks, especially math reasoning and code generation.

I think it can be a cool idea to explore if the findings in the DCD paper generalize to code generation and see if LLMs model the task differently than how humans model it. In fact, we have another paper showing that LLMs do the arithmetic calculations differently than humans: <https://aclanthology.org/2025.emnlp-main.1565.pdf>. We can have a well-scoped project along the line of DCD.

Best,

--Ziyu

From: Mohan Krishna Vallabhaneni <mvallab@gmu.edu>

Date: Tuesday, June 9, 2026 at 19:17

To: Ziyu Yao <ziyuyao@gmu.edu>

Subject: ["Prospective Undergraduate Student"] Interest in LLM Reasoning and Interpretability Research

Hello Professor Yao,

My name is Mohan Krishna Vallabhaneni, and I am a senior Computer Science student at George Mason University. I am writing to ask if there may be an opportunity for me to work with your research group as an undergraduate student researcher.

I read your prospective students page and understood that you expect interested students to read and discuss your publications before reaching out. The paper that interested me the most is:

"Data-driven Circuit Discovery for Interpretability of Language Models"
<https://arxiv.org/pdf/2605.09129v1>

What I understood from this paper is that circuit discovery tries to understand how language models internally solve a task by finding the smaller computational subgraph responsible for that behavior. The part that interested me most is that the paper questions an assumption that I also found important: just because humans define something as one task, it does not mean the model solves every example using one single internal mechanism. The paper shows that small variations in datasets can lead to circuits with low overlap, and that existing methods can sometimes mix different mechanisms into one circuit. I found the idea of Data-driven Circuit Discovery very interesting because it lets the model's internal behavior guide the grouping first, and then discovers separate circuits for each group.

A strength I saw in the paper is that it does not just propose a method directly; it first questions the assumptions behind the existing workflow. I liked that the work connects interpretability with a deeper question: whether our human-defined task boundaries actually match the way a language model organizes computation internally. This made the paper feel very meaningful to me because it is not only about improving a method, but also about asking whether we are explaining the right thing in the first place.

One question I had while reading the paper is how this idea could extend to more open-ended reasoning tasks, especially math reasoning and code generation. In math or programming, a small difference in the problem statement can change the reasoning path completely. I am curious whether data-driven circuit discovery can help reveal when an LLM is using different internal mechanisms for different types of reasoning problems, even when the final task label looks similar to humans.

My research interest is in LLMs for reasoning and planning, especially code generation, math reasoning, and mechanistic interpretability. Since childhood, I have always loved mathematics because I enjoyed understanding how everything connects, from actual numbers all the way to visual representations and more complex ideas. I was taught to deeply understand a subject by imagining how different parts connect in my mind, almost like seeing the logic in thin air. What interests me most is how a final complex outcome arises from many smaller connected ideas. To me, nothing is imaginary in such problems; everything has structure, logic, and calculation behind it. This is the same reason I became interested in research. I want to understand things deeply, not just use them at the surface level.

I also came across GTAs at GMU who are pursuing PhDs while I was working as an Undergraduate Teaching Assistant. Through those conversations, I learned more about what a PhD and research path look like at GMU. My future plan is now tilted toward research, although I feel I am a little late because, as an international transfer student, it took me some time to understand how research opportunities and academic paths work here. Still, I am genuinely interested in learning something deeply and contributing to meaningful research.

I would be happy to work as a research assistant under your guidance if there is such an opportunity, but my main motivation is not money. My real interest is to learn, contribute, and understand this research area deeply. Nothing is more valuable than experience and knowledge. If funding is available, it would certainly help me with my living expenses, but it is not mandatory for me to be involved. I would be grateful for the chance to contribute to your group's work in any useful way.

For Fall 2026, I will only have 7 credits, so I can commit around 10–20 hours per week to research depending on the project needs. I have attached my CV and transcript for your review. If you think there may be a possible fit, I would be grateful for the opportunity to meet with you briefly and discuss how I could contribute to your group.

Thank you very much for your time and consideration.

Sincerely,
Mohan Krishna Vallabhaneni
Senior, B.S. Computer Science
George Mason University
GNumber: G01511747